

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Terminal Points and Interface Data - Power Island

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CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Terminal points between IEC and the EPC

Each terminal point is the physical boundary between the IEC supply (power island) and the EPC's balance of plant. Design conditions are the IEC design values of the IEC component; the EPC side shall be designed for at least the same conditions. Nozzle load limits: attachment A (to follow with IEC-ALP-IF-001 Rev A, 8 weeks after NTP).

Tag	Service	IEC side	EPC side	Medium	Size / rating	Design p / T	Operating (SRC)
IF-01	Natural gas to GT	performance gas heater inlet flange	gas from EPC gas receiving/metering station	natural gas	12" ASME CL600 RF	63 barg / 80 degC	20.67 kg/s, 37.0 barg, 25 degC
IF-02	LDO to GT	liquid fuel skid inlet flange	LDO forwarding/treatment system	LDO	6" CL150	16 barg / 60 degC	22.1 kg/s (LDO operation)
IF-03	Demin water for water injection	water injection skid inlet	demin water storage/pumps	demin water	4" CL150	16 barg / 50 degC	22.5 kg/s (LDO operation)
IF-04	Compressor wash water	wash skid inlet	demin water	demin water	2" CL150	10 barg / 50 degC	5 m3 per wash
IF-05	HP steam	HRSG HP SH outlet stop valve, weld end	main steam piping (EPC)	steam	12" (P91) weld end	210 bar(a) / 610 degC	100.16 kg/s, 187.0 bar(a), 600 degC
IF-06	HP steam	ST main stop valve inlet, weld end	main steam piping (EPC)	steam	12" (P91) weld end	200 bar(a) / 605 degC	180.0 bar(a), 597 degC
IF-07	Cold reheat	HP turbine exhaust, weld end	cold reheat piping (EPC)	steam	24" (P22/A106) weld end	52 bar(a) / 420 degC	40.2 bar(a), 366 degC
IF-08	Cold reheat	HRSG reheater inlet, weld end	cold reheat piping (EPC)	steam	24" weld end	52 bar(a) / 420 degC	39.4 bar(a), 365 degC
IF-09	Hot reheat	HRSG reheater outlet stop valve, weld end	hot reheat piping (EPC)	steam	28" (P91) weld end	50 bar(a) / 610 degC	117.50 kg/s, 37.6 bar(a)
IF-10	Hot reheat	ST IP stop-control valve inlet	hot reheat piping (EPC)	steam	28" (P91) weld end	48 bar(a) / 605 degC	36.0 bar(a), 597 degC
IF-11	LP steam	HRSG LP SH outlet stop valve	LP steam piping (EPC)	steam	20" weld end	10 bar(a) / 330 degC	11.39 kg/s, 5.40 bar(a)
IF-12	LP steam	ST LP admission valve inlet	LP steam piping (EPC)	steam	20" weld end	8 bar(a) / 330 degC	4.95 bar(a)
IF-13	HP / IP / LP bypass valves	valves supplied loose by IEC (incl. desuperheating)	installation in EPC piping, spray water from BFP/CEP (EPC)	steam / water	per valve datasheets	-	start-up / trip
IF-14	Bypass steam to condenser	condenser neck dump devices (IEC)	IP/LP bypass downstream piping (EPC)	steam	2 x 36" + 1 x 28"	16 bar(a) / 250 degC	start-up / trip
IF-15	Condensate from hotwell	hotwell outlet nozzle	CEP suction piping (EPC)	condensate	20" CL150	full vacuum - 3.5 bar(a) / 80 degC	128.89 kg/s, 26.3 degC
IF-16	Condensate to CPH	HRSG CPH inlet	condensate line (EPC)	condensate	12" CL300	25 bar(a) / 180 degC	141.37 kg/s
IF-17	BFP suction	HRSG LP drum downcomer outlet nozzle	BFP suction piping (EPC)	feedwater	16" CL150	10 bar(a) / 185 degC	129.98 kg/s, 158.1 degC
IF-18	HP feedwater	HRSG HP ECO1 inlet	HP feedwater piping (EPC)	feedwater	10" CL2500	235 bar(a) / 200 degC	100.16 kg/s, 209 bar(a)
IF-19	IP feedwater	HRSG IP ECO inlet	IP feedwater piping (EPC)	feedwater	6" CL600	60 bar(a) / 200 degC	29.82 kg/s, 46 bar(a)
IF-20	FGH water return	performance gas heater water outlet	return line to condensate (EPC)	feedwater	4" CL600	60 bar(a) / 260 degC	12.48 kg/s, 75 degC
IF-21	HRSG blowdown	continuous / intermittent blowdown valves outlet	blowdown tanks (EPC)	water	2" / 3"	210 bar(a) / 370 degC	0 in normal operation
IF-22	HRSG drains and vents	drain/vent valves outlet (IEC)	drain collection, flash tanks, silencers (EPC)	water/steam	various	-	start-up
IF-23	Chemical dosing	dosing nozzles on drums/condensate	dosing skids and lines (EPC)	chemicals	1"	-	-
IF-24	Sampling	sample nozzles with root valves (IEC)	sample lines, coolers, SWAS (EPC)	water/steam	1/2"	-	-
IF-25	Aqueous ammonia	ammonia injection skid inlet (IEC)	storage, unloading, forwarding pumps (EPC)	NH4OH 24.5 %	1"	10 barg / 40 degC	125 kg/h
IF-26	Flue gas to stack	HRSG outlet transition flange	stack (EPC), CEMS (EPC)	flue gas	D 9.2 m	+/- 50 mbar / 200 degC	796.7 kg/s, 80.5 degC
IF-27	Circulating water	condenser water box flanges (2 inlets, 2 outlets)	CW pipes (EPC)	seawater	4 x DN1800	4 barg / 40 degC	36,050 m3/h
IF-28	Closed cooling water	IEC coolers inlet/outlet flanges	CCW system (EPC)	demin water + inhibitor	various	10 barg / 60 degC	see IEC-ALP-REQ-001 table 5.1
IF-29	Instrument and service air	IEC skids	compressed air system (EPC)	air	1" - 2"	10 barg	see REQ-001 6.1
IF-30	Nitrogen	HRSG lay-up / generator purge connections	N2 supply (EPC)	N2	1"	10 barg	lay-up
IF-31	Hydrogen / CO2	GT generator gas control panel inlet	bottle racks and piping (EPC)	H2 / CO2	1/2"	16 barg	12 Nm3/day H2
IF-32	Auxiliary steam	gland steam header inlet (IEC)	auxiliary boiler steam (EPC)	steam	4"	16 bar(a) / 320 degC	4.5 t/h at start-up
IF-33	Fire protection	GT enclosure CO2 system (IEC)	fire detection/alarm, transformer deluge, hydrants (EPC)	-	-	-	signals to EPC fire alarm panel
IF-34	GT GSU HV	GT GSU HV bushings	380 kV connection to GIS (EPC)	electrical	400 kV	-	560 MVA
IF-35	ST GSU HV	ST GSU HV bushings	380 kV connection to GIS (EPC)	electrical	400 kV	-	270 MVA
IF-36	UAT supply	IPB tap-off flange (GT and ST IPB)	UAT and MV busduct (EPC)	electrical	21 kV / 15.75 kV	-	-
IF-37	MV supplies	SFC isolation transformer, IEC MV consumers	10.5 kV switchgear feeders (EPC)	electrical	10.5 kV	-	SFC 8.5 MW at start
IF-38	LV supplies	IEC MCCs incoming	0.4 kV switchgear feeders (EPC)	electrical	0.4 kV	-	see REQ-001 7.1
IF-39	DC and UPS	TCS, protection, emergency oil pumps	220 V DC and 230 V AC UPS (EPC)	electrical	-	-	see REQ-001 7.2

Tag	Service	IEC side	EPC side	Medium	Size / rating	Design p / T	Operating (SRC)
IF-40	Control system interface	IEC turbine control system (TCS) gateway	plant DCS (EPC)	data	OPC UA + Modbus TCP, redundant; hardwired trips	-	about 2,500 signals
IF-41	Protection interface	generator/GSU protection panels	switchyard protection and TSO SCADA (EPC)	signals	IEC 61850 + hardwired	-	-
IF-42	Earthing	equipment earthing bosses	earthing grid (EPC)	electrical	-	-	-
IF-43	Foundations	anchor bolts, sole plates, load data (IEC)	foundation design and construction (EPC)	civil	-	-	see REQ-001 section 9